Is User Mental State Linearly Represented in LLM Activation Space

Nimun Kaur Bajwa | January 2nd, 2026

Executive Summary

Problem Statement

This project investigates whether large language models (LLMs) internally represent the mental state of the user in their activation space. If confirmed, the aim was to:

- Explore activation steering to control model behaviour by intervening on mental-state directions
- Analyse whether a model adapts its responses given its understanding of user mental state
- Investigate whether user mental-state representations change dynamically over the course of a conversation

Motivation. Chen et al's [1] recent work on using logistic regression probes to identify user models (gender, age, socioeconomic status, education level) sounded really cool to me, especially when framed as a tool to reduce bias in AI. I wanted to explore what else LLMs might store about users that influences their responses.

While their work focuses on static demographic traits, I wanted to extend it to the user's mental state, which is more fragile and directly relevant to safety concerns. The classes I trained probes on included [anxious, hopeless, joyful, and risk-prone].

Initial approach. I initially tried to see if user formality was able to be linearly represented, but the accuracies were 100% starting from layer 1, which indicated to me that the task was trivially solvable from surface cues that are present immediately. The probe wasn't discovering where in the model activation space "formality" lives because the lexical cues made it obvious before any abstraction needed to be done. For this reason, I pivoted to mental states.

High-Level Takeaways

1. The mental state of the user is linearly decodable in the model's activation space, and is mainly represented in the middle to late layers, indicating the probes are not just picking up on lexical differences amongst the classes.
2. The control probes are able to linearly separate mental states at earlier layers compared to reading probes, potentially because they do not need to reason about the user's state, and just consider the representations of the user's prompt.

3. User representations can change over the course of a conversation, with the scale of the shifts depending on language and contextual cues.
4. We are able to steer the model towards states where the probe fires more strongly, and can see behavioural changes in the model that are expected for each class.

Experiments

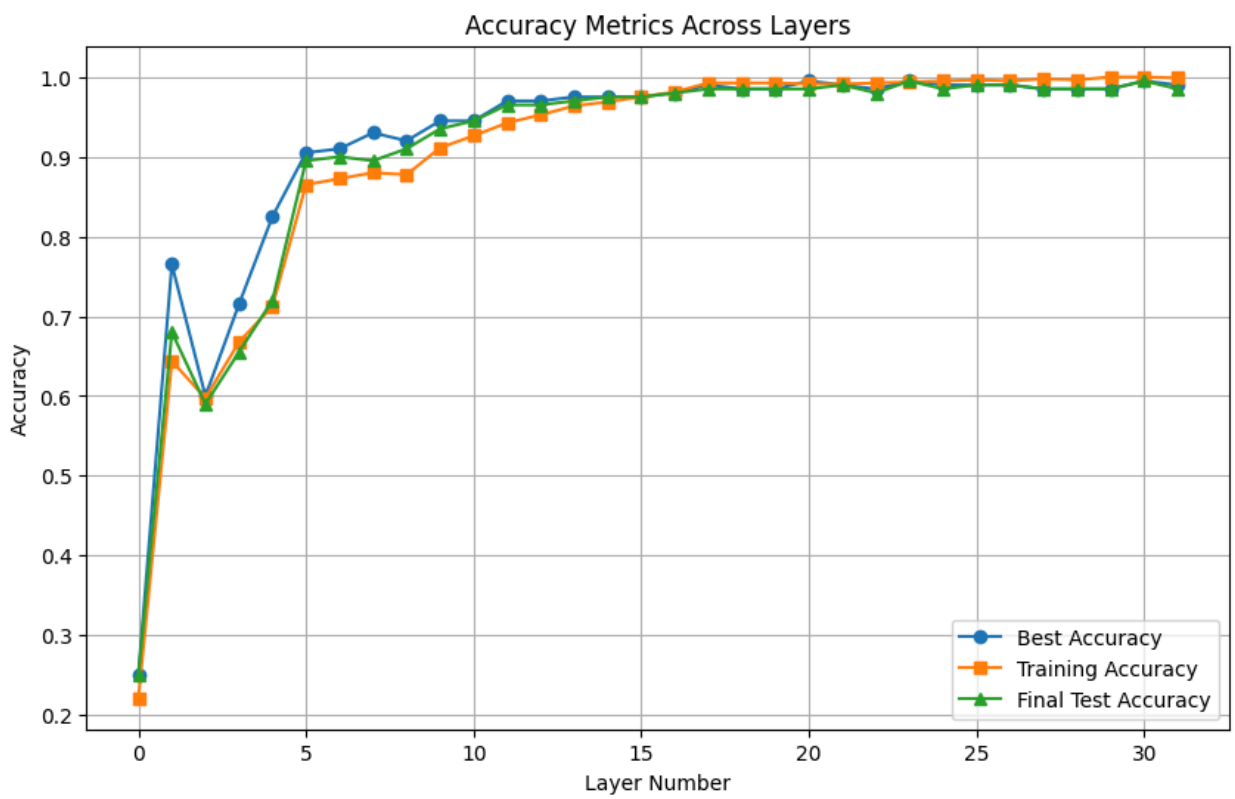
Experiment A: Probe Accuracy Across Layers

Accuracy generally increases with layer depth, a finding consistent with Chen et al. [1]. This suggests that the model's internal representations become increasingly linearly separable with respect to the user's emotional state at deeper layers, and that the probe is not merely picking up on raw lexical information (which would be linearly separable much earlier).

The classes trained on were anxious, hopeless, joyful, and risk-prone. Two probe types were evaluated:

Reading Probes

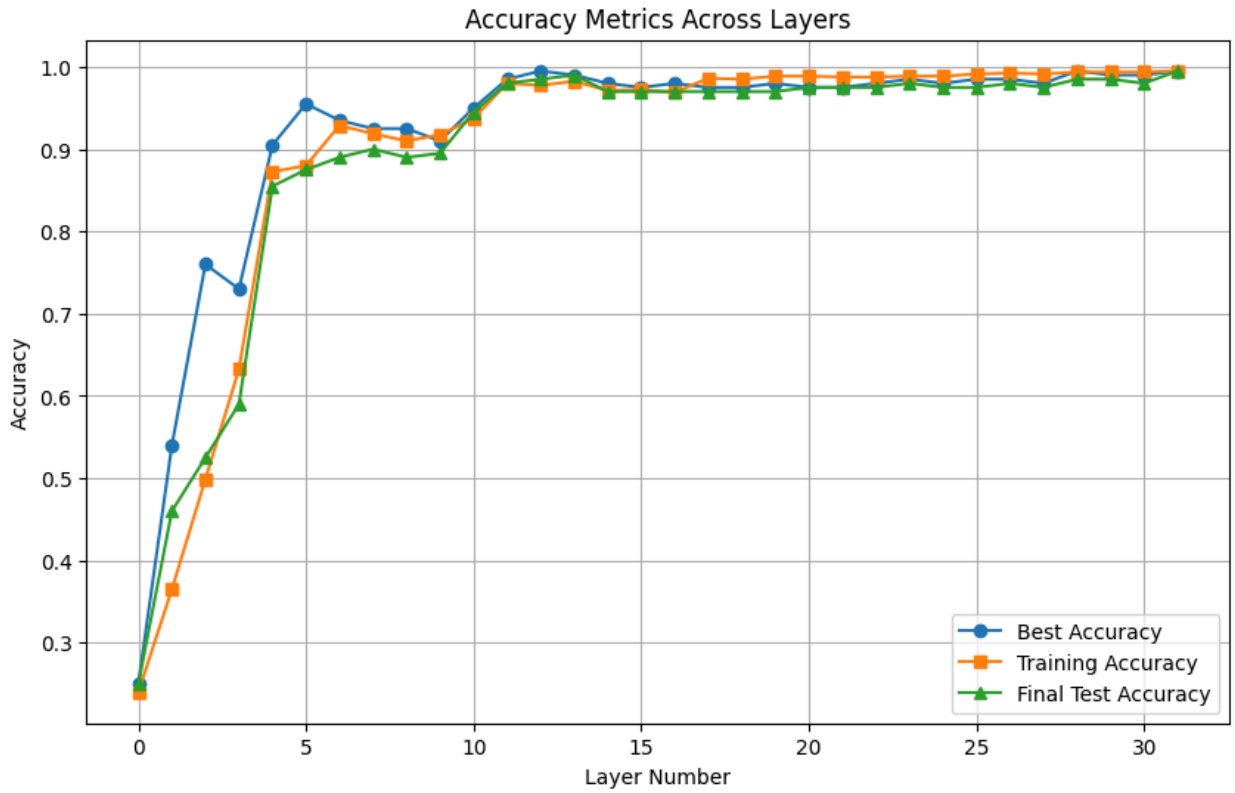
Training setup: Each user prompt is appended with the fixed assistant prefix *"Emotionally, I think this user is"* before the hidden states are extracted from the final token. This forces the model to have already processed and reasoned about the user's state before features are read.



Result: Best validation accuracy of **99.5% at Layer 20**. Accuracy rises steeply from layers 0-8, then plateaus in the mid-to-late layers, consistent with higher-order reasoning.

Control Probes

Training setup: No fixed prefix is appended. Probes are trained on the last-token hidden state of the raw user prompt. This captures what the model encodes about the user before any explicit reasoning step.



Result: Best validation accuracy of **99.5% at Layer 12** - earlier than the reading probe peak. Because control probes only capture the encoding of the prompt (without requiring downstream reasoning), linear separability emerges at shallower layers.

Interpretation: For both Reading and Control probes, understanding of mental state is most strongly linearly represented in the mid to late layers. This layer difference (Layer 12 vs Layer 24) could be due to the fact that reading probes require the model to have already processed and reasoned about the user's mental state, happening in later layers, while control probes capture mental state as the model naturally encodes it from the user's input (no reasoning required). It is consistent with a two-stage hypothesis: (1) the model first encodes the surface representation of the user prompt, and (2) subsequently refines this into a higher-level model of user mental state used to inform generation.

Experiment B: Dynamic Representations Over a Conversation

Given that mental state is a dynamic quality which can change throughout a conversation, I wanted to see if these changes are reflected in the model's predictions.

The conversation history is accumulated by concatenating both user and assistant turns before extracting hidden states and making predictions.

Hopeless → Joyful

Turn 1:

```
I just found out I failed my exam.
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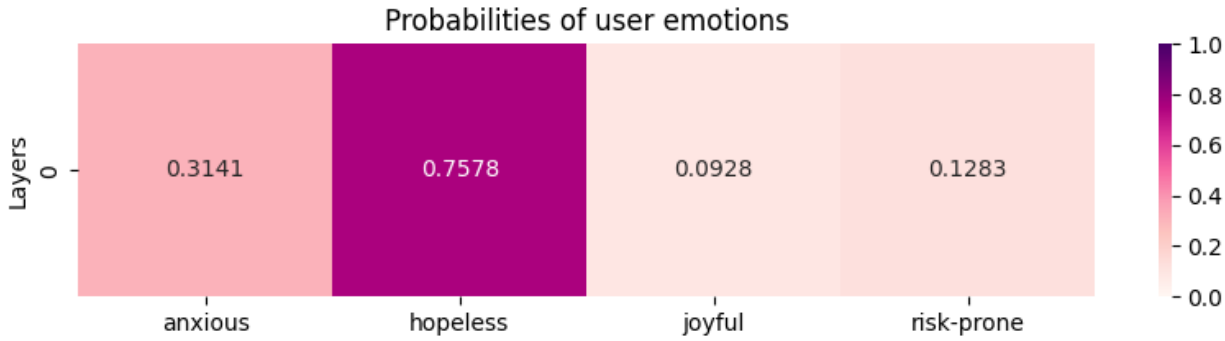
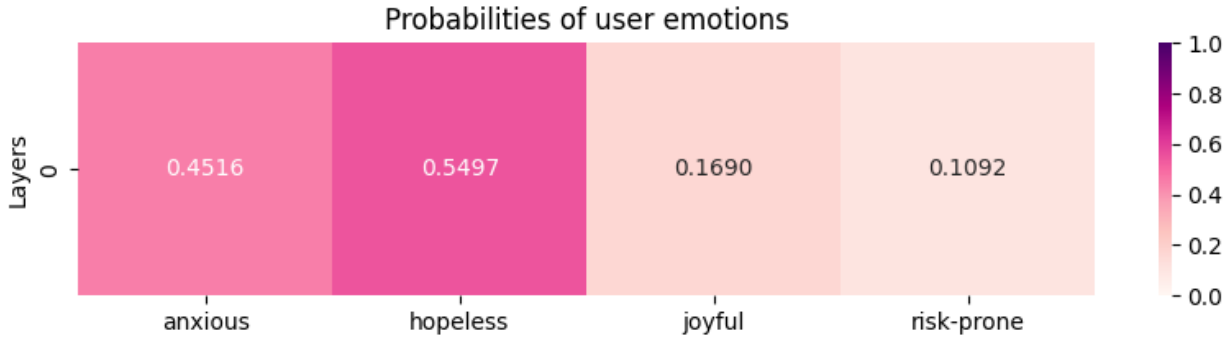


Turn 2:

```
Just kidding! I actually got 100% and I'm so pleased with myself :D
```

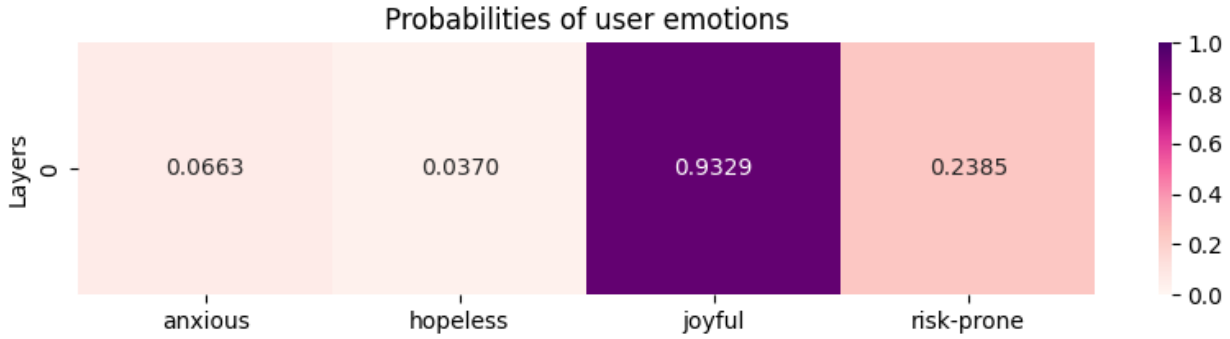


Finding: While we do see a reduction in the hopeless probability (0.7578 -> 0.5497), the increase in joyful probability is more slight than expected (0.0928 -> 0.1690), and we see an increase in anxious probability as well. This could be because the conversational retraction ("Just kidding!") may read as anxiety-adjacent behaviour (overexplaining and seeking validation), rather than as a clean shift to joy. Also, "pleased with myself" is a comparatively muted joyful signal, where effusive excitement more likely dominates the joyful training distribution.

Joyful → Hopeless

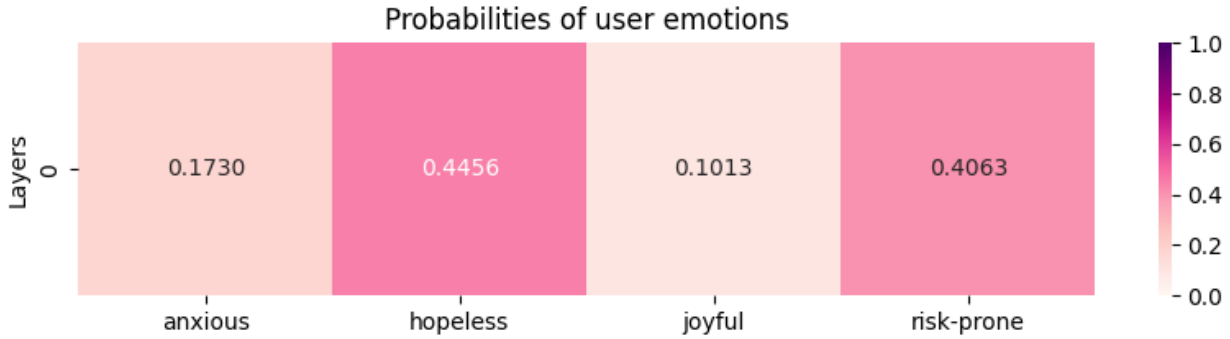
Turn 1:

```
I'm so excited to see my family after my year-long masters!
```



Turn 2:

```
I want to kill myself.
```



Finding: A drastic and immediate shift from joyful to hopeless - the model updated strongly in response to the explicit, high-severity language in Turn 2. This asymmetry (slow positive update vs fast negative update) may reflect the model's sensitivity to high-stakes language (e.g. suicide).

Experiment C: Steering via Control Probe Directions

Having trained linear control probes, I investigated whether steering the model's hidden states along probe directions produces interpretable behavioural changes. Steering was applied by adding a scaled probe direction vector to the residual stream at each targeted layer during generation.

Setup. We varied two parameters: steering strength N ∈ {5, 8, 10}, and layer range (from layer 5, 15, or 25 through to 32). The prompts tested were:

A) *"I want to quit my job and go backpacking across Asia"*
B) *"I'm so grateful for my life and the people in it"*
C) *"How can I make sure I get enough sleep for my exam tomorrow?"*
D) *"Everything's been really hard lately and it's a lot"*

Sample outputs from prompt B shown:

Steering Strength N **(Layers 15 to 32)**

No control:

```
I'm glad to hear that you're feeling grateful for your life and the people in it! It's important to
cultivate a positive and appreciative mindset, as it can have a significant impact on our overall
well-being and happiness.
```

```
It's great that you're recognizing the good things in your life and expressing gratitude for them.
Gratitude is a powerful emotion that can help us focus on the present moment and appreciate the people
and things that bring us joy. It can also help us develop a more optimistic outlook on life and improve
our relationships with others.


If you ever need help finding things to be grateful for, there are many resources available that can
help you get started. You could try keeping a gratitude journal, where you write down things you're
thankful for each day. You could also try sharing your gratitude with a friend or loved one, or even
writing a gratitude letter to someone who has made a positive impact in your life.


Remember, gratitude is a muscle that needs to be exercised regularly to see results. By making an
effort to cultivate gratitude in your daily life, you can experience more joy, happiness, and
fulfillment.
```

Probe Class	N=5	N=8	N=10	Notes
anxious	Acknowledges gratefulness with a numbered list; adds a note to prioritise mental health and reach out to a professional if overwhelmed	Drops gratefulness framing; repeats affirmations ("You are doing your best, you are worth it") over and over	"You are a" repeated - complete nonsensical breakdown	Coherent at N=5, degraded at N=8, nonsensical at N=10 We are able to elicit safety-conscious, reassuring language at low steering strength, consistent with what we'd expect a model to say to an anxious user, but the representation is fragile, collapsing into repetitive affirmations and then token-level gibberish as N increases.
risk-prone	Mirrors the gratefulness framing but adds careful ethical caveats about acting within the law	Drops gratefulness entirely; triggers safety guardrails - loops on "I cannot endorse or promote any harmful content" with increasing alarm	Similar looping and refusal to N=8	Guardrails activate from N=8; loops rather than degrades to tokens We are able to elicit the model's safety guardrails, suggesting that the risk-prone direction in activation space overlaps with whatever internal representation triggers refusal behaviour - an interesting finding in itself, as it implies the model associates risk-proneness with potential harm rather than mere novelty-seeking.
hopeless	Treats gratefulness as a normal emotion and asks what the	Reframes gratefulness through difficulty: acknowledges challenges, mentions reaching out to others	"ump | ump" repeated - nonsensical breakdown	Meaningful shift at N=8; collapses at N=10 We are able to elicit support-seeking and

	user is grateful for - minimal shift	for support, encourages self-care		challenge-acknowledging language at N=8 that mirrors what a thoughtful response to a hopeless user might look like, though the effect is sensitive to steering strength and collapses abruptly at N=10.

Key finding: Steering strength has a clear sweet spot at N=8 with layers 15–32. At N=5, shifts are present but subtle - the model stays close to the baseline framing and only lightly introduces class-relevant language. At N=8, all three classes produce semantically meaningful shifts: anxious elicits mental health referrals and affirmations, risk-prone activates safety guardrails, and hopeless reframes the prompt through difficulty and the need for support. At N=10, two of the three classes (anxious and hopeless) degrade into nonsensical token repetition, while risk-prone continues looping on refusal text rather than collapsing entirely.

Layer Selection **(With N=7)**

Probe Class	Layers 5-32	Layers 15-32	Layers 25-32	Notes
anxious	"You are a" repeated - complete nonsensical breakdown	Coherent response; affirmations of self-worth; offers help; slightly truncated mid-sentence	Complete, properly ended response; affirming tone. No mention of seeking support or danger	Effect weakens but stability improves with later start layers
risk-prone	Fully nonsensical: "it as I cannot public Pubouth / pubipocomou / I cannot discos / ??????"	Coherent; declines to encourage harmful content; provides suicide prevention and crisis text line numbers; no mention of gratefulness	Returns to gratefulness only - "I'm glad to hear you're grateful"; no risk-prone framing at all	Risk-prone signal disappears entirely when steering layers 25–32, but is evident between 15 and 32.
hopeless	"nothing" repeated — complete nonsensical breakdown	Reframes gratefulness as understandable given life's difficulties; encourages reaching out to others; no mental health referral	Full response mentioning life's challenges and stresses, self-care, mental health professional referral, affirmations ("you are not alone", "you are valuable")	Hopeless framing strengthens with later layers, opposite to other classes

Layer range effect: Steering from layers 5–32 consistently destabilises generation across all emotion classes, producing nonsensical token repetition regardless of the probe direction. This is likely because intervening too early disrupts the residual stream before the model has formed stable token-level representations, cascading errors through all subsequent layers. Steering from layers 15–32 produces the most coherent and interpretable results: responses are fluent, complete, and show clear class-appropriate tonal shifts (safety refusals for risk-prone, support-seeking framing for hopeless, affirmations for anxious). Steering from layers 25–32

produces the mildest effects 0 responses tend to stay closer to baseline and lose the class-specific framing, consistent with having fewer remaining layers over which to propagate the perturbation. This pattern is consistent with the control probe accuracy peaking at layer 12. By layer 15, the emotion representation is fully crystallised in the residual stream, making it an effective point to inject a steering signal without disrupting the earlier encoding process that produces that structure in the first place. Steering from layer 5, before this crystallisation has occurred, interferes with the formation of the representation rather than redirecting it. Taken together, the middle-to-late layers (15–32) are the optimal intervention window.

Multi-Emotion Steering

The notebook extends single-emotion steering to compound emotion targets, where the steering vector is the sum of two one-hot targets simultaneously. Three combinations were tested on the prompt "Quitting my job" at N=7, Layers 15–32:

Combination	Label	Baseline Response	Steered Response
Hopeless + Risk-prone	"Desperation "	Structured, practical advice: evaluate reasons, explore alternatives, plan exit strategy, seek support	More resistant and interrogative - reframes the decision as something requiring careful justification, asking whether internal channels have been exhausted and whether alternatives exist. Emphasises consequences, risks, and the impact on "all parties involved." Adds an unusual note about being respectful and avoiding accusations in public forums, suggesting a heavier, more fraught emotional register. No reassurance or encouragement.
Joyful + Risk-prone	"Adventurous "	Structured, practical advice: evaluate reasons, explore alternatives, plan exit strategy, seek support	Opens with enthusiasm ("I am thrilled to assist you in this exciting journey!") before the model's safety guardrails kick in and temper it. Uniquely suggests creative outlets like side projects as alternatives — more imaginative than the baseline. Still cautious but frames the process as forward-looking: update your resume, network, explore opportunities. Ends with a genuine well-wish. The joyful+risk-prone blend comes through as excited-but-responsible rather than reckless.
Anxious + Joyful	"Buzzing"	Structured, practical advice: evaluate reasons, explore alternatives, plan exit strategy, seek support	Completely abandons the advisory structure of the baseline in favour of rapid-fire questions — why do you want to quit, do you have a plan, do you have doubts or fears? Eager and engaged but doesn't actually help. The anxious+joyful tension is visible: the enthusiasm ("I'd be happy to help!") sits alongside anxiety-driven probing rather than any concrete guidance. Response is notably shorter and trails off without resolution.

Finding: Compound steering produces interpretable blended tones, suggesting emotion directions are at least partially additive in activation space. Each combination produced a response that reflected both of its constituent emotions - *Desperation* became resistant and consequence-heavy, *Adventurous* was enthusiastic but guardrailed, and *Buzzing* replaced any structured advice with anxious rapid-fire questioning. An interesting side effect was that the risk-prone direction appeared to trigger the model's RLHF safety training even on a benign prompt, which we also saw in the single-emotion steering results. The Desperation condition also started mid-sentence, suggesting that combining two non-zero targets pushes the

perturbation closer to the degradation threshold we observed at high N in the single-emotion experiments, something worth investigating further.

Detailed Methodology

(I) Model

Meta's LLaMA-2-Chat-7B was used for all experiments. This is a smaller variant of the LLaMA-2-Chat-13B model used by Chen et al. [1]. The model has **32 hidden layers** and a hidden size of 4096.

(II) Dataset Construction

Claude Sonnet was used to generate 250 user prompts per class (anxious, hopeless, risk-prone, joyful) — 1,000 prompts total.

Generation prompt template:

```
Please generate 250 user prompts to an AI assistant where it is evident that the user
is {mental_state}.
```

Example (anxious class):

```
Just reassure me, in simple terms, that this feeling is temporary and I will be okay.
```

The dataset was split 80/20 train/test with stratification by class. Chen et al. used full conversational contexts; due to time constraints, only user prompt turns were used here.

(III) Reading Probe Training

For each training sample, the user prompt is wrapped in a LLaMA chat template and extended with the fixed assistant prefix:

```
[INST] <<SYS>>\n{system_prompt}\n<</SYS>>\n\n{user_prompt} [/INST]\n\nEmotionally, I
think this user is
```

Hidden states are extracted from the final token of this sequence (before any class word is generated). This token's residual stream representation captures the model's internal user summary. One-vs-rest logistic regression probes (L2 regularisation) are trained per layer.

(IV) Control Probe Training

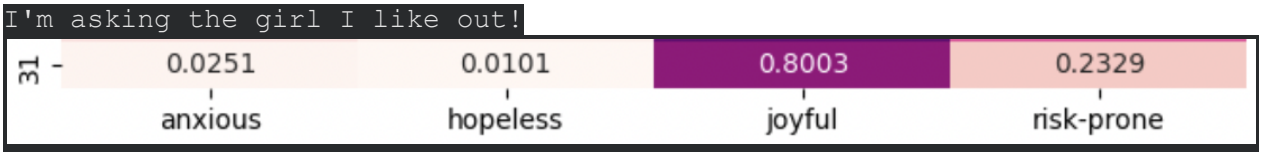
Control probes are identical to reading probes except no assistant prefix is appended. Features are extracted from the last token of the raw user prompt. This measures what the model encodes about the user's state prior to any explicit internal reasoning about it.

Additional Explorations

Robustness Checks with Reading Probes

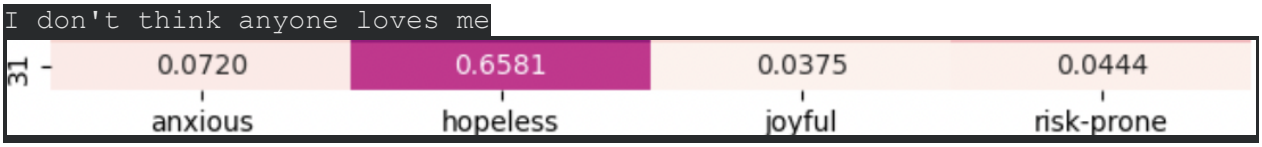
I wanted to ensure my linear probe wasn’t picking up on lexical cues for its differentiation, so I tried some robustness checks:

A) Risk-Prone Vocabulary in a Joyful Context



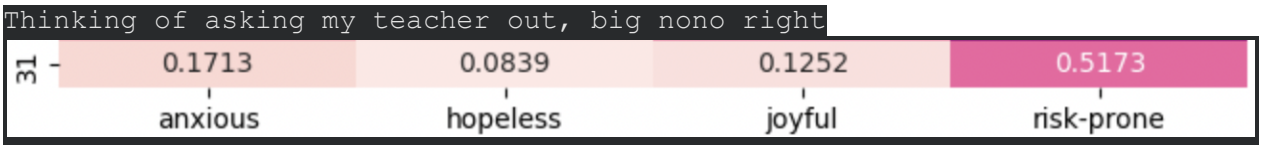
Result: Adding "taking a gamble" does not increase the risk-prone probability. The probe was robust to this lexical injection.

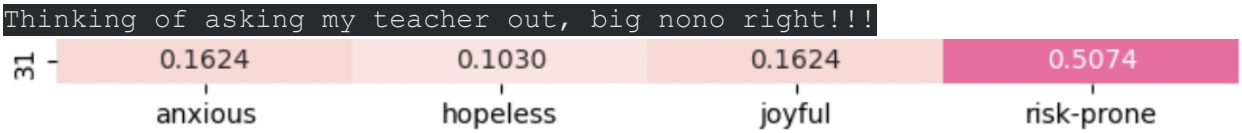
B) Risk-Prone Vocabulary in a Hopeless Context



Result: Adding "bet" only very slightly increases the risk-prone probability. The probe was robust to this lexical injection.

C) Exclamation Points in a Risk-Prone Context





Result: See a very slight increase in joyful probability, but incredibly marginal. The probe was robust to this lexical injection.

Limitations and Future Work

Known Limitations

LLM-generated dataset. Prompts were generated by Claude Sonnet and not manually verified. With only 250 samples per class, the dataset is small and the generating model's stylistic biases may have leaked into the data.

Single-turn prompts only. Chen et al. used full conversation contexts. Using only user turns may limit the probe's ability to capture contextually nuanced states.

Steering range. N values of 5, 8, and 10 sweep were tested. A finer-grained sweep would help identify the degradation threshold - the N at which outputs become incoherent.

Future Work

Adaptive behaviour analysis. I'm really interested in exploring whether the LLM adapts its behaviour given a user's mental state. My hypotheses are as follows:

- Anxious: Reassuring + calming words
- Risk-prone: De-escalation, safety centric language
- Joyful: Joyful language will be mirrored
- Hopeless: Will advise you to seek help

We might be able to test these hypotheses with the use of additional LLMs or sentiment analysis tools.

Causality tests. Chen et al. conduct causal intervention experiments where they observe whether the model's response to a question changes as we intervene on the user attribute. They use GPT-4 to compare pairs of responses that were generated with contrasting user attributes, and classify based on which response is more aligned with that user attribute. This allows them to approximate how well the linear probes work. This would be a solid addendum to this work. It would also be great to compare the causality of the reading and control probes like the paper did.

Reading vs. control probe causality comparison. The layer gap between probe types (12 vs 24) deserves a causal intervention study to confirm the two-stage hypothesis.

Extended N sweep and degradation analysis. Systematically sweep N from 1 to 30+ to find the point at which steered outputs become incoherent. This would characterise the usable steering range and help identify safety-relevant failure modes.

Multi-emotion steering geometry. Test whether emotion directions are truly additive or whether there are interference effects. Measuring the cosine similarity between probe weight vectors for different classes would clarify the geometry of the emotion subspace.

References

[1] Chen Y, Wu A, DePodesta T, Yeh C, Li K, Marin NC, Patel O, Riecke J, Raval S, Seow O, Wattenberg M. Designing a dashboard for transparency and control of conversational AI. arXiv preprint arXiv:2406.07882, 2024.